

^{60}Ga $\varepsilon+\beta^+$ decay (69.4 ms) 2001Ma96,2021Or01,2025Ow01

Parent: ^{60}Ga : $E=0$; $J^\pi=2^+$; $T_{1/2}=69.4$ ms 2; $Q(\varepsilon+\beta^+)=14160$ 15; $\% \varepsilon+\beta^+$ decay=100.0

$^{60}\text{Ga}-J^\pi$: from the Adopted Levels of ^{60}Ga .

$^{60}\text{Ga}-T_{1/2}$: unweighted average of 70 ms 15 (2001Ma96), 70 ms 13 (2002Lo13), 76 ms 3 (2017Ku12), 70.8 ms 20 (2017GoZT,2020Gi02), 69.4 ms 2 (2021Or01), and 57.7 ms 98 (2025Ow01).

$^{60}\text{Ga}-Q(\varepsilon+\beta^+)$: deduced by evaluators using ^{60}Zn mass excess of -54174.5 5 (2021Wa16) and ^{60}Ga mass excess of -40015 15; weighted average of -40016 15 (2021Or01), -40005 30 (2021Pa44), and -40034 46 (2023Wa10).

$^{60}\text{Ga}-\% \varepsilon+\beta^+$ decay: $\%(\varepsilon+\beta^+)p=1.6$ 7, $\%(\varepsilon+\beta^+)\alpha<0.023$ 20 (2001Ma96).

2001Ma96: ^{60}Ga was produced via the $^{28}\text{Si}(^{36}\text{Ar},p3n)$ fusion evaporation reaction using a 4.71 MeV/u ^{36}Ar primary beam at the GSI On-Line Mass Separator. A 55-keV $^{60}\text{Ga}^{1+}$ beam was extracted and implanted into a tape for $\beta\gamma$ measurements and carbon foils for βp and $\beta\alpha$ measurements in a beam-on/beam-off mode. Positrons were detected using a plastic scintillator and γ rays were detected using Ge detectors. Particles were detected using Si ΔE -E telescopes at another beam line. Measured E_γ , I_γ , $\gamma\gamma$ -coin, $\beta\gamma$ -coin, $E_p(>0.6$ MeV), I_p , E_α , and I_α . Deduced levels, J , π , ^{60}Ga $T_{1/2}$, and ^{60}Ga mass using the known ^{60}Zn mass, the E_γ from the $T=1$ IAS in ^{60}Zn , and the Coulomb displacement energy systematics (1997An07). Identified a total of 802 proton events and deduced $\%(\varepsilon+\beta^+)p=1.6$ 7. Identified 9 α candidate events and deduced $\%(\varepsilon+\beta^+)\alpha<0.023$ 20.

2021Or01: ^{60}Ga was produced via the projectile fragmentation of a 345-MeV/u ^{78}Kr primary beam impinging on a 5-mm thick ^9Be target, identified in the RIKEN BigRIPS separator by $B\rho$ - ΔE -ToF, transported to the exit of the ZeroDegree spectrometer, and implanted into the Wide-range Active Silicon Strip Stopper Array for Beta and ion detection (WAS3ABi) consisting of three 1-mm-thick DSSDs. γ rays were detected using the EUroball-RIKEN Cluster Array of 12 cluster-type Ge detectors. Measured $E_p(>1.5$ MeV), E_γ , I_γ , $p\gamma$ -coin, $\beta\gamma$ -coin, $\gamma\gamma$ -coin, and implant-decay time correlations. Deduced levels, ^{60}Ga $T_{1/2}$, and ^{60}Ga mass using the known ^{59}Zn mass, the E_p and E_γ from the $T=2$ IAS in ^{60}Ga populated by ^{60}Ge decay in the same experiment.

2025Ow01: ^{60}Ga was produced via the projectile fragmentation of a 150-MeV/nucleon $^{78}\text{Kr}^{36+}$ primary beam impinging on a 517-mg/cm² thick ^9Be target, filtered by the NSCL A1900 separator and the radio frequency fragment separator, identified using the ΔE -ToF method, and implanted into a 1-mm-thick DSSD with 16×16 1.2-mm wide strips at the center of the Summing NaI (SuN) detector. Beam ions and decay positrons were detected using the DSSD and γ rays were detected using the SuN detector. Measured implant-decay correlations, γ -ray total absorption spectra (TAS), sum of segments spectra (SoS), and multiplicity spectra. Deduced β -feeding intensities, Gamow-Teller strength distributions, and $T_{1/2}$ of parent ^{60}Ga . Compared with shell-model calculations using GXPF1A and KB3G Hamiltonians.

 ^{60}Zn Levels

$E(\text{level})^\dagger$	J^π	Comments
0	0^+	
1003.53 10	2^+	
2193		
2558.54 23	(2^+)	
3035		
3200		
3510		
3627		
3710		
3808		
3812		
3972		
4180		
4200		
4351		
4400		
4776		
4851.97 32	2^+	$T=1$, isobaric analog state of ^{60}Ga g.s.
4913		
5200		
≈5460		
≈5733		
≈6020		
≈6321		

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^{60}Ga $\varepsilon+\beta^+$ decay (69.4 ms) [2001Ma96,2021Or01,2025Ow01](#) (continued) ^{60}Zn Levels (continued)

$E(\text{level})^\dagger$	$E(\text{level})^\dagger$	$E(\text{level})^\dagger$
≈ 6637	≈ 7683	≈ 8894
≈ 6969	≈ 8067	≈ 9338
≈ 7317	≈ 8470	≈ 9805

† From a least-squares fit to γ -ray energies for levels connected with γ transitions. For other levels below 5.3 MeV, from [2025Ow01](#) adopted from the RIPL-3 library, originally from the Adopted Levels of ^{60}Zn ([2013Br15](#)) rounded to the nearest keV. For levels above 5.3 MeV, $E(\text{level})$ is a sample of calculated continuum-state energies with energy spacing (273 keV) that follows the energy resolution of the SuN detector.

 ε, β^+ radiations

$E(\text{decay})$	$E(\text{level})$	$I\beta^+ \uparrow @$	$I\varepsilon \uparrow @$	$\text{Log } ft^\dagger$	$I(\varepsilon+\beta^+)^\ddagger @$	Comments
(4.36×10^3)	≈ 9805	0.49	0.01	3.8	0.5	$I(\varepsilon+\beta^+)$: 0.5 +13-4.
(4.82×10^3)	≈ 9338	1.18	0.02	3.7	1.2	$I(\varepsilon+\beta^+)$: 1.2 +9-12.
(5.27×10^3)	≈ 8894	0.10	9.077×10^{-4}	5.0	0.1	$I(\varepsilon+\beta^+)$: 0.1 +4-1.
(5.69×10^3)	≈ 8470	1.69	0.01	3.9	1.7	$I(\varepsilon+\beta^+)$: 1.7 +13-11.
(6.09×10^3)	≈ 8067				$<0.1^\#$	
(6.48×10^3)	≈ 7683				$<0.1^\#$	$I(\varepsilon+\beta^+)$: 0.0 +3-0.
(6.84×10^3)	≈ 7317				$<0.1^\#$	
(7.19×10^3)	≈ 6969				$<0.1^\#$	
(7.52×10^3)	≈ 6637	0.10	2.641×10^{-4}	5.8	0.1	$I(\varepsilon+\beta^+)$: 0.1 +9-1.
(7.84×10^3)	≈ 6321				$<0.1^\#$	
(8.14×10^3)	≈ 6020	1.00	0.002	5.0	1.0	$I(\varepsilon+\beta^+)$: 1.0 +74-10.
(8.43×10^3)	≈ 5733	4.89	0.01	4.4	4.9	$I(\varepsilon+\beta^+)$: 4.9 +13-49.
(8.70×10^3)	≈ 5460	3.09	0.005	4.7	3.1	$I(\varepsilon+\beta^+)$: 3.1 +21-8.
(8960 15)	5200	4.69	0.01	4.5	4.7	$I(\varepsilon+\beta^+)$: 4.7 +15-47.
(9247 15)	4913	11.0	0.01	4.2	11.0	$I(\varepsilon+\beta^+)$: 11.0 +22-41.
(9308 15)	4851.97	37.7	0.05	3.7	37.7	$I(\varepsilon+\beta^+)$: 37.7 +75-47 (2025Ow01); $I(\varepsilon+\beta^+)=44.1$ 32 (2021Or01).
(9384 15)	4776				$<0.1^\#$	
(9760 15)	4400				$<0.1^\#$	
(9809 15)	4351				$<0.1^\#$	
(9960 15)	4200				$<0.1^\#$	
(9980 15)	4180				$<0.1^\#$	
(10188 15)	3972	8.59	0.01	4.6	8.6	$I(\varepsilon+\beta^+)$: 8.6 +28-16.
(10348 15)	3812	5.30	0.005	4.8	5.3	$I(\varepsilon+\beta^+)$: 5.3 +14-52.
(10352 15)	3808				$<0.1^\#$	
(10450 15)	3710				$<0.1^\#$	
(10533 15)	3627	5.79	0.005	4.8	5.8	$I(\varepsilon+\beta^+)$: 5.8 +11-12.
(10650 15)	3510	2.10	0.002	5.3	2.1	$I(\varepsilon+\beta^+)$: 2.1 +6-7.
(10960 15)	3200				$<0.1^\#$	
(11125 15)	3035	0.40	3.017×10^{-4}	6.1	0.4	$I(\varepsilon+\beta^+)$: 0.4 +3-4.
(11602 15)	2558.54	7.79	0.005	4.9	7.8	$I(\varepsilon+\beta^+)$: 7.8 +10-12 (2025Ow01); $I(\varepsilon+\beta^+)=9.2$ 10 (2021Or01).
(11967 15)	2193				$<0.1^\#$	
(13157 15)	1003.53	0.2		6.8	0.2	$I(\varepsilon+\beta^+)$: 0.2 +13-2 (2025Ow01); $I(\varepsilon+\beta^+)=17$ 5 (2021Or01).
(14160 15)	0	3.8 6	0.0027 4	8.3 1	3.8 6	

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^{60}Ga $\varepsilon+\beta^+$ decay (69.4 ms) **2001Ma96,2021Or01,2025Ow01** (continued) ε, β^+ radiations (continued)

[†] Calculated by evaluators using the BetaShape code (2023Mo21).

[‡] From 2025Ow01. $I(\varepsilon+\beta^+)$ at each E(level) is extracted by simultaneously fitting the simulated TAS, SoS, and multiplicity spectra to data via the χ^2 minimization. The sum is normalized to $\Sigma I(\varepsilon+\beta^+)=100$. The decay scheme built via β -delayed γ -ray spectroscopy (2001Ma96,2021Or01) is considered incomplete due to a large gap of 9 MeV between the highest observed level at E=4852 and $Q(\varepsilon)=14160$ 15. There should be missing transitions from unobserved levels in the gap. Therefore, the $\varepsilon+\beta^+$ -feeding values from γ +ce intensity balance at each level are considered upper limits due to the incomplete decay scheme.

[#] $I(\varepsilon+\beta^+)$ values below 0.1 were set to 0.0 in 2025Ow01 due to sensitivity limitations.

[@] Absolute intensity per 100 decays.

 $\gamma(^{60}\text{Zn})$

I_γ normalization: absolute γ -ray intensities per 100 decays of ^{60}Ga were measured by 2021Or01 based on the total number of implanted ^{60}Ga (7.6×10^5) and $\beta\gamma$ detection efficiencies, correcting for DAQ dead time.

E_γ	I_γ [†]	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Comments
^x 669.3 3	0.58 15					% I_γ =0.36 9
^x 850.8 1	1.21 16					% I_γ =0.75 10
^x 913.9 3	0.48 16					% I_γ =0.30 10
1003.5 1	100 5	1003.53	2 ⁺	0	0 ⁺	% I_γ =62.0 31 E_γ : from 2021Or01. Other: 1003.7 2 (2001Ma96). I_γ : from 2021Or01. Other: 100 17 (2001Ma96).
^x 1028.6 2	0.61 13					% I_γ =0.38 8
^x 1188.4 1	2.58 16					% I_γ =1.60 10
^x 1201.8 2	0.47 11					% I_γ =0.29 7
^x 1413.7 2	0.57 11					% I_γ =0.35 7
^x 1442.1 1	0.65 13					% I_γ =0.40 8
^x 1481.4 1	2.10 16					% I_γ =1.30 10
1554.7 3	11.3 8	2558.54	(2 ⁺)	1003.53	2 ⁺	% I_γ =7.0 5 E_γ : from 2021Or01. Other: 1554.9 6 (2001Ma96). I_γ : from 2021Or01. Other: 12 5 (2001Ma96).
^x 1780.8 6	0.32 16					% I_γ =0.20 10
^x 2047.2 4	1.13 32					% I_γ =0.70 20
2293.2 4	10.2 8	4851.97	2 ⁺	2558.54	(2 ⁺)	% I_γ =6.3 5 E_γ : from 2021Or01. Other: 2293.0 10 (2001Ma96). I_γ : from 2021Or01. Other: 10 5 (2001Ma96).
^x 2334.2 3	1.29 32					% I_γ =0.80 20
^x 2434.2 2	2.90 32					% I_γ =1.80 20
2558.8 4	13.7 10	2558.54	(2 ⁺)	0	0 ⁺	% I_γ =8.5 6 E_γ : weighted average of 2559.0 8 (2001Ma96) and 2558.7 4 (2021Or01). I_γ : from 2021Or01. Other: 13 5 (2001Ma96).
^x 2624.3 5	0.48 16					% I_γ =0.30 10
^x 2826.0 2	2.10 32					% I_γ =1.30 20
^x 2884.0 4	1.29 32					% I_γ =0.80 20
^x 2996.8 2	3.2 5					% I_γ =1.98 31
^x 3337.4 1	11.5 10					% I_γ =7.1 6
^x 3394.8 1	11.3 10					% I_γ =7.0 6
3848.5 4	61 5	4851.97	2 ⁺	1003.53	2 ⁺	% I_γ =37.8 31 E_γ : from 2021Or01. Other: 3848.3 7 (2001Ma96). I_γ : from 2021Or01. Other: 57 13 (2001Ma96).
^x 3889.1 3	4.5 13					% I_γ =2.8 8
^x 4000.9 2	4.5 7					% I_γ =2.8 4
^x 4805.0 4	0.65 16					% I_γ =0.40 10

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⁶⁰Ga ε+β⁺ decay (69.4 ms) 2001Ma96,2021Or01,2025Ow01 (continued)

γ(⁶⁰Zn) (continued)

<u>E_γ</u>	<u>I_γ[†]</u>	<u>E_i(level)</u>	<u>Comments</u>
^x 4850.2 5	0.32 16		%I _γ =0.20 10
^x 4891.9 3	0.65 16		%I _γ =0.40 10

[†] For absolute intensity per 100 decays, multiply by 0.62.

^x γ ray not placed in level scheme.

^{60}Ga $\varepsilon + \beta^+$ decay (69.4 ms) 2001Ma96,2021Or01,2025Ow01

Decay Scheme

Legend

Intensities: I_γ per 100 parent decays

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$

